THE EULER CHARACTERISTIC OF A COHERENT MODULE EQUALS THE ALTERNATING SUM OF THE EULER CHARACTERISTICS OF ITS HIGHER DIRECT IMAGES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Coherence and finiteness of higher direct images of a proper morphism. Stacks Project, Cohomology of Schemes, Proposition coherent-proposition-proper-pushforward-coherent (tag 02O5) and Lemma coherent-lemma-quasi-coherence-higher-direct-images (tag 02O6). Makes the alternating sum finite and each term finite dimensional.
- The Leray spectral sequence. Stacks Project, Cohomology, Lemma cohomology-lemma-Leray (tag 01F2). Provides the spectral sequence whose convergence gives the result.

Source reference. Original reference 02O5: sources/stacks-chow/tags.txt, source lines 3504–3504. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XJ: sources/stacks-chow/tags.txt, source lines 2520–2520. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01F2: sources/stacks-chow/tags.txt, source lines 1855–1855. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0BDW: sources/stacks-chow/tags.txt, source lines 14329–14329. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let $f : Y \to X$ be a morphism of proper schemes over k. Let $\mathcal{G}$ be a coherent $\mathcal{O}_Y$-module. Then*

$$\chi(Y, \mathcal{G}) = \sum (-1)^i \chi(X, R^i f_* \mathcal{G})$$

Proof. The formula makes sense: the sheaves $R^i f_* \mathcal{G}$ are coherent and only a finite number of them are nonzero, see Cohomology of Schemes, Proposition 02O5 and Lemma 01XJ. By Cohomology, Lemma 01F2 there is a spectral sequence with

$$E_2^{p,q} = H^p(X, R^q f_* \mathcal{G})$$

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converging to $H^{p+q}(Y, \mathcal{G})$. By finiteness of cohomology on X we see that only a finite number of $E_2^{p,q}$ are nonzero and each $E_2^{p,q}$ is a finite dimensional vector space. It follows that the same is true for $E_r^{p,q}$ for $r \geq 2$ and that

$$\sum (-1)^{p+q} \dim_k E_r^{p,q}$$

is independent of r. Since for r large enough we have $E_r^{p,q} = E_\infty^{p,q}$ and since convergence means there is a filtration on $H^n(Y, \mathcal{G})$ whose graded pieces are $E_\infty^{p,q}$ with $p + q = n$ (this is the meaning of convergence of the spectral sequence), we conclude. Compare also with the more general Homology, Lemma 0BDW. $\square$